

For more on the Immunization Agenda 2030 Framework for Action see <https://www.immunizationagenda2030.org/framework-for-action>

complex epidemiological transitions, macroeconomic issues, and political instability, all of which constrain their resources. Without sufficient and predictable financing, many LMICs will not make any substantial progress towards achieving the goals of the Immunization Agenda 2030 Framework for Action. Failing to meet these goals would have negative effects on the lives and wellbeing of all people.

A *Lancet* Correspondence by Uchenna A Amaechi and colleagues² suggested that countries can further enhance their fiscal space for immunisation by using innovative cofinancing mechanisms. A feasible opportunity that should be urgently explored is national health insurance, a health financing option that many countries are adopting to drive progress towards universal health coverage.³ Although coverage varies, some countries already have substantial pools, with their national health insurance generating revenues through contributions from individuals themselves or through alternative channels for those who are unable to pay.⁴ With the concerted effort of crucial stakeholders, national health insurance can be galvanised to cofinance primary health care, most pertinently, immunisation services, including vaccines. Redirecting pooled funds to immunisation would require the development and implementation of multiple policy reforms with sector-wide implications. This call to action is intended to stimulate researchers, donors, policy makers, and programme implementers to rethink and begin to redesign the financing model for primary health care and immunisation, especially in LMICs, in a manner that positions national health insurance as a potential important funding source.

Our key recommendations for action to cofinance immunisation through national health insurance are: to specify routine immunisation, including vaccines, in the health benefit package of national health insurance schemes; to establish contract agreements with national immunisation programmes

to procure and supply recommended vaccines through existing mechanisms on behalf of children and adults that are covered under national health insurance schemes; to implement a prospective input-and-output-based payment method for immunisation services, including vaccines; and to deploy an enhanced immunisation performance monitoring system to track and link provider incentive payments with health-care facility functionality, immunisation service utilisation, and immunisation service quality.

We declare no competing interests. The views and opinions expressed in this Correspondence are those of the authors only and do not necessarily represent those of their affiliated organisations.

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One Health pandemic preparedness: the role of companion animals in disease transmission

Urbanisation creates novel interfaces that facilitate cross-species

transmission of emerging infectious diseases. Outwardly, so-called city encroachment into natural areas creates periurban interfaces between people, livestock, and wildlife. Inwards, companion animals are fully integrated into urban life, creating so-called animal encroachment into dwellings and novel human-animal interfaces, with the potential for complex urban multi-species transmission networks.

Historically, the urban interface linking humans, companion animals, and urban wildlife has received limited attention in pandemic preparedness and response. However, recent events highlight its importance: several companion dogs and cats were infected with SARS-CoV-2 by their owners, and at least one human COVID-19 outbreak originated from pet hamsters.¹ Most recently, companion cats in Poland and South Korea and dogs in France and Canada were infected with avian influenza H5N1 clade 2.3.4.4b following outbreaks in urban wild and domestic birds.²

WHO, the World Organisation for Animal Health, the Food and Agriculture Organization, and the UN Environment Programme have jointly called for the prioritisation and acceleration of One Health, an integrated, unifying approach that aims to sustainably balance and optimise the health of people, animals, and ecosystems.³ Through this collaboration, they aim to build intersectoral workforces that “strengthen and sustain prevention of pandemics and health threats at source, targeting activities and places that increase the risk of zoonotic spillover between animals to humans”.⁴ Aligned with this call, WHO published the Mosaic Respiratory Surveillance Framework,⁵ highlighting the importance of community event-based surveillance (CEBS) in targeting at-risk populations at the human-animal interface. Although CEBS is deemed essential for rapidly detecting,

reporting, and verifying signals in the community, there is no standard CEBS for emerging or novel viruses affecting humans and their companion animals. We propose closely monitoring these urban human–animal interfaces using standardised One Health disease surveillance within households. Household transmission investigations (HHTIs)⁶ have been successfully implemented to investigate human disease dynamics by collecting high-quality epidemiological, clinical, virological, and serological data in a well defined, closed setting. We propose expanding WHO's HHTI protocol^{5,6} into a One Health HHTI, capturing companion animal data and characterising the interactions at the human–companion animal interface. The phased expansion requires veterinary public health expertise to revise and adapt protocols to include companion animals, develop multi-species case and contact definitions, and determine minimum data requirements for timely public health decision making. Once the One Health HHTI protocol is created, determining processes for specimen collection, analytics, and data governance is essential to ensure the cross-sectoral feasibility and sustainability of the approach. The One Health HHTI protocol is needed to address the full spectrum of disease control necessary for global health security.

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Biodiversity and planetary health: a call for integrated action

In the face of escalating biodiversity loss, the imperative role of comprehensive research and conservation strategies has never been more pressing. The National Biodiversity Future Centre (NBFC) in Italy stands at the forefront of tackling biodiversity loss, pioneering innovative approaches within the Mediterranean's biodiversity hotspot. In this Correspondence, we aim to highlight the pressing need for synchronised efforts in protecting our planet's biological wealth, which is fundamental to sustaining life as we know it.

Biodiversity's influence extends to cultural and mental wellbeing, intertwined with cultural practices and contributing to identity and belonging. The need for cohesive strategies and the innovative, pivotal role of a centre on biodiversity research in the Mediterranean is crucial to advancing our understanding of monitoring,

conservation, and restoration of biodiversity and to drive sustainable biological resource use.

Further research is essential as biodiversity is a cornerstone of wellbeing, providing essential resources and regulating services such as climate moderation, flood mitigation, and nutrient cycling, crucial for maintaining ecological equilibrium.¹ The connection between biodiversity, sustainable development, and planetary health becomes imperative as we face global diseases and environmental crises. Biodiversity loss endangers essential ecological services, affecting billions of people globally.² Environmental degradation and biodiversity loss contribute to a substantial portion of the global disease burden.³ The emergence of infectious diseases underscores the necessity of preserving biodiversity for global health security, stressing the importance of adopting integrated strategies at various levels.⁴

The NBFC, established under the Post-Covid Recovery and Resilience Plan, is at the forefront of this endeavour. It advocates nature-based solutions to harmonise human requirements with biodiversity conservation, ensuring ecosystem services and preserving environmental resources that are essential for human survival and prosperity. The NBFC works in close synergy with institutions and private companies for the protection of the environment and the enhancement of biodiversity. In Italy and the Mediterranean, the rejuvenation of biodiversity is essential for sustainable development. Mediterranean biodiversity, a wealth of bioactive compounds, is crucial for pioneering treatments against non-communicable diseases. Natural substances are key in creating innovative health solutions that blend wellbeing with economic growth.⁵

The conservation of biodiversity is, therefore, not just a matter of environmental stewardship but a critical component in the continuing



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